

**TEAM ASSIGNMENT (100 pts)**

1. In teams of (2), you are to design, build and test a electrically powered vehicle.
 - a. Vehicle should use motor(s), gears, belts, and pulleys as needed from your hardware kits for transmission components and laser cut parts (1/8" plywood) for the frame/housings. 1/8" Plywood is available in the Wondry. Tell them you are in ME2160.
 - i. Rapid prototype parts may be used for extraneous non-essential decorative parts i.e. "Bling". Think cow skull with horns hood ornament.
 - ii. Transmission ratio should probably be **at least** 3:1. i.e. the motor should rotate 3 times for every 1 rotation of the output shaft.
 - iii. Vehicle should fit within a 8" x 8" x 8" box.
 - b. Full credit will be awarded to vehicles that traverse at least 10 feet unaided. Partial credit will be awarded proportionally, i.e. 8 ft = 80%.
 - c. Vehicles should be painted and/or otherwise decorated to highlight the team's unique style.
 - d. Ten addition points will be awarded for each of the following categories.
 - i. Lightest weight.
 - ii. Most school spirit. (Go Does!)
 - iii. Most compact (fits in smallest size by volume).
 - iv. Fastest.
 - v. Highest "Coolness Factor" as voted on by the class.
 - vi. Most "Rube Goldberg" design. (Google it!)
 - vii. Slowest. (This may not be as easy as you think)
 - viii. Pulls the most weight. Can anyone say **Tractor Pull**? Must provide a hole on back to attach string.
2. The Proof of Concept Demonstration is intended to serve as a first public iteration and as an example to use for a design review with TA's and the Instructor. This iteration will be graded qualitatively based on level of effort.